8 Pediatric Otolaryngology

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8.1 Pediatric Airway Evaluation and Management

◆ Key Features

- The airway is relatively narrower and more tenuous in children.
- The potential for airway emergency is high.
- Many conditions causing respiratory distress in infants resolve spontaneously with growth.

The pediatric airway is proportionally smaller than that of the adult: the tongue is relatively larger and more anterior, the soft palate descends lower, the adenoid is larger, the epiglottis is omega-shaped, larger, and more acutely angled toward the glottis; the cricoid ring is narrower, the trachea is shorter and narrower, the surrounding soft tissue is looser, and cartilaginous structures are less rigid. Thus the pediatric airway is more prone to compromise by infection, inflammation, neoplasia, and normal breathing.
Foreign body aspiration may be a life-threatening emergency. An aspirated solid or semisolid object may lodge in the airway. If the object is large enough to cause nearly complete obstruction of the airway, then asphyxia may rapidly cause death. Infants are at risk for foreign body aspiration because of their tendency to put everything in their mouths and because of immature chewing. Children may be asymptomatic. If present, physical findings may include stridor, fixed wheeze, or diminished breath sounds. If obstruction is severe, cyanosis may occur.

◆ Clinical

Signs and Symptoms

- Stridor: Harsh, high-pitched sound of turbulent airflow past partial obstruction in upper airway
  - Inspiratory stridor signifies supraglottic obstruction.
  - Biphasic stridor signifies glottic or subglottic obstruction.
  - Expiratory stridor signifies tracheal or large bronchial compression.
- Stertor: Low-pitched, snorting sound resulting from partial nasal/nasopharyngeal/hypopharyngeal obstruction
- Wheezing: A continuous whistling or musical sound on expiration from a small bronchiole constriction

For subjective assessment of respiratory distress, see **Table 8.1**. Indications for intubation for airway compromise include P_{AO_2} < 60 mm Hg with F_{IO_2} > 0.6 (without cyanotic heart disease), P_{ACO_2} > 50% (acute, unresponsive to other intervention), actual or impending obstruction, neuromuscular weakness (maximum negative inspiratory pressure over –20 cm H_2O, vital capacity < 12–15 mL/kg), and an absent cough/gag reflex.

Table 8.1 Subjective assessment of respiratory distress

	None	Mild	Moderate	Severe
Stridor	None	Mild	Moderate at rest	Severe on inspiration and expiration or none with markedly decreased air entry
Retractions	None	Mild	Moderate at rest	Severe, marked use of accessory muscles
Color	Normal	Normal	Normal	Dusky or cyanotic
Level of consciousness	Normal	Restless when disturbed	Anxious; agitated; restless when disturbed	Lethargic, depressed

Data from Davis HW, Dartner JC, Galvis AG, et al. Acute upper airway obstruction: croup and epiglottitis. *Pediatr Clin North Am* 1981;28(4):859.

Differential Diagnosis

- Supralaryngeal
  - Adenoid hypertrophy
  - Macroglossia
  - Mass (nasopharyngeal, base of tongue)
  - Choanal atresia
  - Foreign body
  - Cellulitis
  - Neck/pharyngeal abscess
- Laryngeal
  - Laryngomalacia (most common overall cause of pediatric stridor)
  - Vocal fold paralysis (unilateral often is iatrogenic or trauma-related, bilateral is often due to central nervous system [CNS] lesion or dysfunction).
  - Laryngeal web
  - Subglottic stenosis or hemangioma
  - Papillomata
  - Laryngeal cleft
  - Viral croup
  - Epiglottitis
  - Gastric reflux
- Tracheobronchial
  - Tracheomalacia

- Bronchomalacia
- Stenosis
- Vascular ring
- Airway foreign body
- Bronchitis
- Bronchiolitis
- Tracheoesophageal fistula (TEF)

◆ Evaluation

History

A thorough history of respiratory distress (cyanosis, apnea, dyspnea, retractions, grunting) including age of onset, frequency, degree, and rate of progression, ameliorating and exacerbating factors (including positional), as well as feeding difficulties and fevers, should be elicited. Difficult delivery, history of prematurity, and postpartum complications (asphyxia, duration of intubation) should be noted.

Physical Exam

On the physical exam, note respiratory rate, nasal flaring, intercostal and supraclavicular retraction, gasping, or respiratory fatigue. Examine the mouth using a tongue blade, unless epiglottitis is suspected. Auscultate the chest and neck. Note change in breathing when positioning child upright, supine, prone, and on each side. Visualize airway with flexible laryngoscope, unless epiglottitis is suspected. More severe cases may require bronchoscopy and possibly esophagoscopy. Bronchoscopy (rigid or flexible) may be both diagnostic and therapeutic (**see A1** in **Appendix A**).

Imaging

Upright anteroposterior (AP) and lateral soft tissue X-rays of the airway (ideally on full inspiration with head in extension) best evaluate the upper airway. Additionally, AP and lateral chest X-rays can detect extrinsic compression or evaluate for evidence of airway foreign body. If child has feeding difficulties, obtain a barium esophagram. Magnetic resonance imaging (MRI) with MR angiography (MRA) is a good alternative to angiography for vascular malformations. Obtain gastric emptying scans and esophageal pH monitoring if reflux is suspected.

For foreign body aspiration, employ posteroanterior inspiratory and expiratory chest radiography (to look for hyperinflation), computed tomography (CT), or fluoroscopy. Studies may be false-negative; if index of suspicion remains high, bronchoscopy in the OR is indicated.

Other Tests

Although pulmonary function testing with a flow-volume loop can help distinguish inspiratory versus expiratory as well as intrathoracic versus extrathoracic obstruction, the test requires a cooperative subject and is not often feasible in children < 6 years. Polysomnography is very helpful in the evaluation of possible pediatric sleep-related respiratory disorders; a differentiation between obstructive and central apnea can be obtained. This is highly recommended in children with a history of neurologic disorder.

◆ Treatment Options

Acute choking, with respiratory failure associated with airway foreign body obstruction, may be successfully treated at the scene using standard first aid techniques such as the Heimlich maneuver, back blows, and abdominal thrusts. Even in less urgent settings, expeditious removal of airway foreign bodies is recommended and a work-up may be performed.

Medical

Definitive management will of course depend upon the specific diagnosis. But in general terms, for the child with airway compromise, continuous monitoring with pulse oximetry is necessary. Supplemental humidified oxygen, racemic epinephrine, or heliox may be implemented. Systemic steroids are often employed.

Reflux medications such as lansoprazole (Prevacid, Takeda Pharmaceuticals USA, Deerfield, IL) to decrease gastric acid production, as well as metoclopramide (Reglan, Schwarz Pharma Inc., Milwaukee, WI) to promote gastric emptying, can help with reflux irritation of the airway. Upper respiratory infections should be empirically treated as an inpatient with ceftriaxone 75 mg/kg/day intravenously (IV), or comparable antibiotic covering *Streptococcus pneumoniae*, *Streptococcus pyogenes*, and *Haemophilus influenzae*.

Surgical

Again, definitive management will, of course, depend upon the specific diagnosis. In cases of severe compromise, pediatric tracheotomy may be necessary. In laryngomalacia, redundant laryngeal tissue can be excised, and aryepiglottic folds can be divided using endoscopic scissors or CO_2 laser. However, many cases of laryngomalacia may be managed with observation (see **Chapter 8.2**). Subglottic stenosis can be dilated or a cricoid split or cartilage graft reconstruction performed (see **Chapter 8.7**). Obstructing masses should be removed, if possible. Subglottic hemangioma management is discussed in **Chapter 8.14**.

◆ Outcome and Follow-Up

The child should be monitored closely overnight in case of bleeding or edema compromising airway. Oxygen saturation should be monitored. Specific conditions are discussed in detail in the following chapters.

Outcome depends on the diagnosis and management. The child can be followed with the usual well-child checks, and immunizations should be kept up to date.

8.2 Laryngomalacia

◆ Key Features

- Laryngomalacia is the most common cause of stridor in infants (accounts for ~ 75% of infantile stridor).
- Often self-limited; most patients are symptom-free by 12 to 24 months of age.
- Treatment for 90% of cases is expectant observation.
- Its etiology is unknown.

Laryngomalacia is a temporary physiologic dysfunction due to abnormal flaccidity of laryngeal tissues or incoordination of supralaryngeal structures.

◆ Epidemiology

Laryngomalacia is the most common congenital airway abnormality. Overall, laryngomalacia accounts for 60% of cases of chronic laryngeal stridor. It is more common among males than females (2:1). Concomitant airway abnormalities are found in 12 to 37% of patients. Comorbidities, including prematurity, cardiovascular malformation, and neurologic and congenital or chromosomal abnormalities, are present in ~ 41% of patients.

◆ Clinical

Signs and Symptoms

Most commonly, patients present with intermittent inspiratory stridor that is relieved by neck extension and a prone position. Stridor is exacerbated by agitation. In extreme cases, patients become cyanotic, have a poor oral intake, have chest retractions, and develop pectus excavatum.

Differential Diagnosis

Other causes of stridor in the early pediatric age group include:

- Unilateral or bilateral vocal fold paralysis
- Laryngeal cleft
- Choanal atresia
- Airway hemangioma
- Laryngeal web
- Airway foreign body
- Acquired or congenital subglottic stenosis
- Craniofacial anomalies
- Glottic cysts
- Laryngeal reflux

- Saccular cyst
- Tracheomalacia
- Papillomatosis

Acid-reflux disorders (see **Chapter 5.5**) have been documented in up to 80% of patients with laryngomalacia.

◆ Evaluation

Physical Exam

An examination of any child with a possible breathing problem should discern whether there is an oxygenation problem, and if so, an oxygen requirement. In the physical examination, one should assess for the location of a possible obstruction and include auscultation, inspection for chest retraction, assessment for cyanosis and other anomalies such as micrognathia. To diagnose laryngomalacia and assess for other upper airway abnormalities, direct flexible endoscopic examination during respiration must be performed. Inward collapse of supraglottic structures is visualized on inspiration. The vocal folds are normal in appearance and motility.

Direct laryngoscopy and bronchoscopy in the operating room is the definitive evaluation.

Imaging

Radiologic examination is not generally necessary. However, if there is concern about dysphagia, barium swallow is helpful to assess esophageal function and to assess for evidence of vascular rings or other obstructive lesions or evidence of tracheoesophageal fistula (TEF).

Pathology

Histologically, submucosal edema and lymphatic dilatation are seen. Mechanisms of obstruction have been described by numerous authors, some of whom considered the occurrence of two or more synchronously as causal in airway obstruction. These factors include an inward collapse of aryepiglottic folds, an elongated epiglottis (flaccid) curled on itself, anterior and medial collapsing movements of the arytenoid cartilages, posterior and inferior displacement of the epiglottis, short aryepiglottic folds, and an overly acute angle of epiglottis.

◆ Treatment Options

Medical

If an infant has good progress, which is indicated by adequate weight gain and normal development, then surgical therapy is not necessary. Instead, supportive care can be the mainstay of treatment.

Surgical

In one series of 985 patients with laryngomalacia, 12% required surgical intervention. Patients who should be considered for surgical management are those with severe stridor and failure to thrive, obstructive apnea, weight loss, severe chest deformity, cyanotic attacks, pulmonary hypertension, or cor pulmonale. Supraglottoplasty is performed using carbon dioxide laser or laryngeal microscissors, or other cold instruments such as pediatric ethmoid through-cutting forceps. Most commonly, surgery involves removal of the prolapsing aryepiglottic fold with cuneiform cartilage or division of a tight, short aryepiglottic fold. Unilateral supraglottoplasty has been advocated by some to reduce the risk of supraglottic stenosis. If the epiglottis is displaced posteriorly, an epiglottopexy can be performed.

Potential complications include continued airway obstruction and posterior stenosis. In the event of continued airway obstruction, a tracheotomy may be necessary until the child “outgrows” laryngomalacia. The use of postoperative antibiotics has not been well evaluated in the literature and is controversial. Antireflux medications are used routinely.

◆ Outcome and Follow-Up

Supraglottoplasty relieves symptoms of airway obstruction in 90% of patients. Most infants require only a short hospital stay (1–3 days).

8.3 Bilateral Vocal Fold Paralysis

◆ Key Features

- Bilateral vocal fold paralysis is the second most common cause of infantile stridor.
- It typically requires a tracheotomy for maintenance of airway until vocal fold mobility returns or definitive airway surgery is performed.
- Acquired causes are most common, even in infants.

Bilateral vocal fold paralysis (BVFP) is a potentially lethal problem requiring aggressive management, typically tracheotomy, at least in the short term. Depending on the cause, spontaneous recovery can occur. Recovery, however, is generally a slow process, taking up to a year. A variety of surgical approaches to widen the airway have been described. Usually, however, the voice quality is degraded when there is an intervention to enlarge/improve the laryngeal airway.

◆ Epidemiology

Twenty-five percent of BVFP is congenital. Most of the remaining cases are late presentations of central lesions. Acquired cases are most likely to occur as a result of surgery in the chest or from forceps delivery or infections.

◆ Clinical

Signs

Vocal fold immobility can be seen using flexible laryngoscopy in the clinic. Stridor is a sign of BVFP.

Symptoms

- Stridor
- Typically a normal voice, episodic respiratory distress (e.g., with upper respiratory tract infections)
- Weak cough; aspiration if underlying cause is a neural lesion above the nodose ganglion (inferior ganglion of vagus nerve)

Differential Diagnosis

- Vocal fold fixation
- Laryngeal mass
- Laryngeal web

◆ Evaluation

Physical Exam

Assess for stridor, respiratory effort and rate, color, and weight (plot on a growth chart). Flexible laryngoscopy is mandatory: it may show twitching of cords with respiration; it typically shows paramedian vocal folds.

Imaging

Magnetic resonance imaging (MRI) should scan from skull base to mediastinum (to image the complete course of laryngeal nerves).

Other Tests

Tests depend on the clinical scenario. For example, consider audiometry if a central nervous system (CNS) disorder is present; consider a biopsy of nonvascular laryngeal mass lesion, if present.

Pathology

Congenital Causes

- Laryngeal anomalies, malformations
- CNS anomalies: Arnold-Chiari malformation, hydrocephalus, encephalocele, leukodystrophy, cerebral dysgenesis
- Peripheral neuropathy: neonatal myasthenia gravis, benign congenital hypotonia, Werdnig-Hoffmann's disease, Charcot-Marie-Tooth's disease, arthrogryposis, viral neuropathy
- Birth trauma, including perinatal hypoxia

Acquired Causes

- Sequelae of cardiac or esophageal surgery
- Neoplasia
- Infections
- Trauma

◆ Treatment Options

Medical

No medical treatment options are available other than supportive care initially to maintain ventilation and oxygenation during definitive treatment planning.

Surgical

- Tracheotomy and monitoring for spontaneous recovery (minimum of 1 year)
- Arytenoidopexy, endoscopic vocal fold lateralization procedures
  - Lateralization sutures passed through the vocal process and thyroid ala lateralize the vocal fold without mucosal destruction
- Arytenoidectomy
  - External approach (lateral cervical or via laryngofissure)
  - CO_2 laser via endoscopic approach
- Posterior cartilage graft
- Laryngeal reinnervation: not widely used in children

◆ Complications

Accidental tracheotomy displacement can be fatal in the early recovery period. Other complications related to laryngeal airway surgeries include dysphonia, aspiration in 4 to 6%, and dyspnea in 3 to 8% of patients undergoing arytenoid procedures. Other procedures have not been studied enough to ascertain their complication rates. Assessment for aspiration should be done preoperatively, as posterior glottic expansion surgery can increase risk of aspiration.

◆ Outcome and Follow-Up

Initially, the patient is monitored in an intensive care unit (ICU) setting with continuous pulse oximetry. Placement of “stay sutures” for any pediatric tracheotomy is mandatory. These two polypropylene (Prolene, Johnson & Johnson, Inc., Cincinnati, OH) sutures, placed through cartilage adjacent to the tracheal opening at the time of surgery and secured to the neck skin with Steri-Strips (3M, St. Paul, MN), greatly facilitate the replacement of a tracheotomy tube into the airway if there is accidental displacement. To help prevent displacement, the tracheotomy appliance should be secured to the skin with four sutures as well as the umbilical necktie.

Overall, arytenoidopexy and arytenoidectomy yield high rates of successful decannulation. Such patients require long-term follow-up.

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8.7 Subglottic Stenosis

◆ Key Features

- The subglottis is the narrowest part of the infant airway.
- Acquired subglottic stenosis is most commonly found; the most common related factor is intubation.
- Subglottic stenosis of 70% or more of the lumen is associated with daily symptoms.

The subglottis, the narrowest part of the pediatric airway, is composed of a complete cartilaginous ring, the cricoid, and loose submucosa that swells when irritated (as in, for example, croup). It is the region most likely to be affected by pressure from a too-large or frequently moving endotracheal tube. Congenital subglottic stenosis most commonly results from an abnormally shaped cricoid, with intraluminal lateral shelves, resulting in an oval shape to the lumen. Intervention is individualized to the patient: sometimes a watch-and-wait approach works; other children benefit from some type of surgery.

◆ Epidemiology

Incidence is ~ 1.5 cases per million, but it occurs in 1 to 8% of neonates requiring intubation.

◆ Clinical

An abnormal, narrow airway causes stridor. It is, therefore, worth reviewing the typical dimensions of the “normal” airway (**Table 8.2**).

Table 8.2 Normal airway size by age

Age	Normal subglottic airway diameter (mm)	Expected endotracheal tube size	Expected bronchoscope size
Premature	3.5–4.5	2.5–3.0	2.5
0–3 months	5.0	3.5	3.0
3–9 months	5.5	4.0	3.5
9–24 months	6.0	4.5	4.0
2–4 years	6.5–7.0	5.0	4.0
4–6 years	7.5	5.5	5.0
6–8 years	8.0	6.0	6.0

Used with permission from Van de Water TR, Staecker H, eds. *Otolaryngology: Basic Science and Clinical Review*. Stuttgart/New York: Thieme; 2006:213.

Signs

The signs include stridor, respiratory distress, and typically a normal voice.

Symptoms

- Unsuccessful extubation of a patient in the neonatal intensive care unit (NICU)
- Stridor; may be biphasic, may present only with agitation
- Respiratory distress exacerbated by upper respiratory tract infection
- Recurrent croup
- Feeding problems
- Slow growth, failure to thrive

Differential Diagnosis

Other causes of stridor with normal voice include laryngomalacia, subglottic cyst, subglottic hemangioma, and tracheal stenosis.

◆ Evaluation

Physical Exam

Flexible laryngoscopy should be done to assess vocal fold movement. The subglottis is sometimes seen with this exam, but the scope should not be passed through the glottis because of the risk of inducing vasovagal reflexes. Bronchoscopy will show subglottic stenosis. The degree of narrowing and the length of the narrowed segment are important to measure.

The Cotton-Meyer grading system apparently correlates to symptoms and prognosis (**Table 8.3**). Grade I stenoses typically are asymptomatic unless there is an upper respiratory tract infection.

Table 8.3 Cotton-Meyer grading system for subglottic stenosis

Grade	Degree of narrowing
I	< 50%
II	50–70%
III	71–99%
IV	Total obstruction

Data from Myer CM, O'Connor DM, Cotton RT. Proposed grading system for subglottic stenosis based on endotracheal tube size. *Ann Otol Rhinol Laryngol* 1994: 108; 319.

Imaging

A soft tissue lateral X-ray, airway fluoroscopy, or computed tomography (CT) may show the subglottic anatomy. Note that airway fluoroscopy involves significant radiation exposure.

Other Tests

A complete evaluation for reflux (including a barium swallow, a gastric scintiscan, or a pH probe) may be helpful. Pulmonary function tests and video stroboscopy are indicated in cooperative older patients.

Pathology

Congenital subglottic stenosis may be cartilaginous or membranous (a thickened submucosa). The subglottis is the narrowest part of the pediatric airway in normal circumstances, so pressure necrosis from an endotracheal tube is more likely here than elsewhere in the airway. Acquired stenosis may be caused by mucosal necrosis with healing by granulation tissue and subsequent fibrosis, although deeper injuries including cartilage necrosis can occur. Factors related to injury include size of endotracheal tube, number of reintubations, tube movement, and tube material (polyvinyl chloride is considered safest).

◆ Treatment Options

Medical

Antireflux medications have a role in management. If an acid-reflux disorder is not diagnosed with traditional tests, prophylactic reflux medications should be given, especially if there is a suggestion of reflux laryngitis on examination.

Surgical

Dilation

Dilation is an option for mild soft stenosis.

Laser Division

Laser division (CO_2, argon, or potassium titanyl phosphate [KTP] laser) is an option for early stenosis (granulation tissue), crescent-shaped bands, and thin circumferential webs.

Cricoid Split

A cricoid split is indicated in neonates with solitary subglottic stenosis and failure to extubate. Traditionally, the criteria for the procedure are weight at least 1,500 g, no ventilator support for 10 days prior to repair, oxygen requirement less than 30%, no congestive heart failure in the month prior to the repair, no acute upper respiratory infection, and no antihypertensive medications required in the 10 days prior to extubation. Basically, the approach is like a tracheotomy, but the anterior midline incision in the trachea extends through the cricoid and the inferior part of the thyroid cartilage. An endotracheal tube is placed such that the incision is open by 2 to 3 mm and left in situ for 7 to 14 days, and steroids are used prior to extubation.

Laryngotracheoplasty

If more than 3 mm of circumference gain is needed, a cartilage graft is necessary. Auricular or costal cartilage may be used. Grafts can be placed anterior and posterior in the subglottis. Stenting is required if the posterior cricoid is divided; this may be achieved by endotracheal intubation for 1 to 2 weeks.

Only mature stenoses should be reconstructed with an open procedure. Cricotracheal resection is reserved for amenable airway lesions.

◆ Complications

Emergency complications can involve airway obstruction or respiratory problems. Causes include mucous plugs, granulation tissue, aspiration of stenting materials (if used), hematoma, hemorrhage into the airway, or pneumothorax. Treatment is aimed at the underlying problem. Treatment may require airway suctioning for mucous plugs; use of aerosolized steroids (e.g., dexamethasone 0.25 to 1.0 mg/kg per day to a maximum of 20 mg/d) to reduce granulations; chest tube placement for pneumothorax, hemorrhage, or hematoma; or return to the operating room for bronchoscopic or possible open treatments.

Other complications include cricoid split failure (repeat intubation for 72 additional hours with a half-size smaller tube; if that fails, then tracheotomy) and laryngotracheoplasty failure (infection may lead to graft necrosis, treated by antibiotics and revision surgery).

◆ Outcome and Follow-Up

A postoperative chest radiograph should always be obtained for open reconstructive procedures. Continuous O_2 saturation monitoring should be overnight; an intensive care unit (ICU) setting should be available for any open procedures. Sedation is required, but paralysis is not desired, so that in the event of accidental extubation, the child can make breathing efforts. Also, spontaneous ventilation is preferred because of improved airway clearance of secretions and decreased muscle weakness that can develop from not breathing for 7 to 10 days.

Humidification and chest physiotherapy are helpful. Racemic epinephrine may be used to treat postextubation edema. Steroids are judiciously avoided in patients with new cartilage grafts, as steroids may inhibit mucosalization and compromise neovascularity. Only one dose is given the morning of extubation.

Antireflux medications are typically employed. Prophylactic postoperative antibiotics are typically given. Management by a skilled pediatric intensivist is required, to reduce potential morbidities.

These children require long-term follow-up. Depending on the procedure employed and the patient's symptoms, serial bronchoscopy may be indicated.

8.10 Diseases of the Adenoids and Palatine Tonsils

8.10.1 Adenotonsillitis

◆ Key Features

- Adenotonsillitis is most commonly viral in etiology.
- Bacterial etiology is similar to acute otitis media (OM):
  - Group A β-hemolytic *Streptococcus pneumoniae*
  - *Moraxella catarrhalis*
  - *Haemophilus influenzae*
- Chronic infection is typically polymicrobial.
- There is growing evidence for the role of biofilm.

Acute adenotonsillitis most commonly presents in children 5 to 10 years of age and young adults 15 to 25 years of age. The primary consideration in its accurate diagnosis and treatment is prevention of secondary complications particularly associated with group A β-hemolytic *Streptococcus pneumoniae*, including rheumatic fever and poststreptococcal glomerulonephritis. Suppurative complications avoided by early and appropriate management include peritonsillar abscess and deep neck space infection.

◆ Epidemiology

The average incidence of all acute upper respiratory infections is five to seven per child per year. It is estimated that children have one streptococcal infection every 4 to 5 years. Group A *Streptococcus* is isolated in 30.0 to 36.8% of children with pharyngitis, and asymptomatic carriage of group A *Streptococcus* is ~ 10.9% for children aged 14 or younger.

◆ Clinical

Signs

- Erythremic, exudative palatine tonsils
- Tonsilloliths
- Trismus
- Cervical adenopathy
- Palatal petechiae (infectious mononucleosis)

Symptoms

- Halitosis
- Sore throat
- Odynophagia

- Purulent rhinorrhea
- Postnasal drip
- Nasal obstruction
- Fever

Differential Diagnosis

- Adenotonsillar hypertrophy
- Acute pharyngitis (bacterial or viral)
- Peritonsillar abscess
- Infectious mononucleosis
- Lymphoma, leukemia, or other neoplasm (unilateral/asymmetric tonsillar enlargement)

◆ Evaluation

Physical Exam

In the head and neck exam, focus on examination of the oral cavity, inspection of palatine tonsils looking for enlargement, erythema, peritonsillar cellulitis, abscess, and exudates. Palpation of the neck is performed to assess for cervical adenopathy.

Imaging

Contrast-enhanced computed tomography (CT) if concerned about retropharyngeal or deep neck space infection.

Labs

Complete blood count (CBC) with differential; monospot, if clinically indicated.

Other Tests

Do a throat swab for culture and sensitivity and latex agglutination tests for group A β-hemolytic *Streptococcus.*

◆ Treatment Options

Medical

Analgesics and antipyretics are prescribed.

Relevant Pharmacology

Antibiotic therapies include:

- **Amoxicillin:** Penicillin-based β-lactam antibiotic with bactericidal action due to interference with bacterial cell wall synthesis
  - 45 mg/kg per day divided every 8 hours for 7 to 10 days
- **Amoxicillin + Clavulanate:** Clavulanate has itself little antibacterial action, but it is a potent β-lactamase inhibitor, protecting the penicillin-based antibacterial action.

- 45 mg/kg per day (amoxicillin component) divided every 12 hours for 7 to 10 days
- **Azithromycin:** Macrolide, semisynthetic derivative of erythromycin. Bactericidal action is through inhibition of protein synthesis via binding to the 50S ribosomal RNA subunit.
  - 10 mg/kg for 1 dose, then 5 mg/kg per day for 5 days
- **Cefuroxime axetil:** Second-generation cephalosporins' bactericidal action is due to the inhibition of peptidoglycan synthesis interfering with cell wall synthesis, similar to penicillin's.
  - 30 mg/kg per day divided every 12 hours for 7 to 10 days

Surgical

Surgery entails a delayed tonsillectomy and/or adenoidectomy for recurrent disease (**Table 8.5**). Currently, it is rare to perform a tonsillectomy in the setting of an acute infection (quinsy tonsillectomy). The presence of a peritonsillar abscess requires acute incision and drainage; removal of the tonsil to allow drainage is rarely needed.

Table 8.5 Adenotonsillectomy indications and contraindications

Absolute indications • Complication of sleep apnea secondary to tonsillar hypertrophy (i.e., cor pulmonale) • Suspected tonsil malignancy • Febrile convulsions secondary to tonsillitis • Tonsillar hemorrhage • Severe failure to thrive with enlarged tonsils
Relative indications • Obstructive sleep apnea • Recurrent acute tonsillitis • 5–7/yr for 1 yr • 5/yr for 2 yr • 3/yr for 3 yr • > 2 weeks of school missed over 1 yr • Peritonsillar abscess, persistent or recurrent • Chronic tonsillitis (throat pain, halitosis, cervical adenitis) • Severe odynophagia • Tonsillolithiasis (if severe, persistent) • Acquired dental or orofacial abnormalities • Chronic carrier of *Streptococcus* • Recurrent/chronic otitis media (adenoidectomy alone)
Contraindications • Cleft palate* • Velopharyngeal insufficiency* • Bleeding diathesis (unless correctible medically; e.g., von Willebrand patients can undergo surgery with appropriate treatment perioperatively)

*Relative contraindication for tonsillectomy, absolute contraindication for adenoidectomy (although partial adenoidectomy may be considered).

◆ Complications

See **Chapter 8.10.2** for a detailed discussion of complications. Primary tonsillar hemorrhage often requires operative management (0.5 to 2.2 per 100). Secondary (delayed) tonsillar hemorrhage is often managed conservatively with close observation (0.1 to 3 per 100).

◆ Outcome and Follow-Up

In the absence of complications in an otherwise healthy child, adenotonsillectomy is a same-day procedure with analgesia and possibly postoperative antibiotics depending on the technique and the surgeon's preference. Hospital admission criteria are also discussed in **Chapter 8.10.2**. Follow-up is 3 to 6 weeks after tonsillectomy and adenoidectomy.

8.10.2 Adenotonsillar Hypertrophy

◆ Key Features

- Adenoids and palatine tonsils are a common source of upper airway obstruction in childhood.
- Peak size is achieved by ~ 5 years of age.
- Airway compromise presents as obstructive sleep apnea (OSA).

Adenotonsillar hypertrophy leads to a spectrum of airway obstruction in children and is a common indication for surgery in the pediatric age group. Adenoids and palatine tonsils enlarge over the first to fifth years of life. Progressive involution occurs by 12 to 18 years of age, though the course is highly variable and is influenced by both allergy and the frequency of recurrent tonsillar and upper respiratory tract infections.

◆ Epidemiology

Adenotonsillar hypertrophy is the most common cause of sleep-related upper airway obstruction in children. Forty percent of children snore; the incidence of true obstructive apnea is estimated to be 3%.

◆ Clinical

Signs

- Mouth open posture
- Failure to thrive and feeding difficulties
- Hyponasal speech
- Adenoid facies (high arched palate, flattened midface, "allergic shiners," mouth-open posture)

- Behavioral disturbances
- Daytime somnolence

Symptoms

- Snoring and/or apneic pauses
- Restless sleep
- Choking or gagging noises while asleep
- Hyperactivity, attention deficit symptoms
- Enuresis
- Nasal obstruction

Differential Diagnosis

- Acute or chronic tonsillitis, adenoiditis (bacterial or viral)
- Acute pharyngitis (bacterial or viral)
- Peritonsillar abscess
- Infectious mononucleosis
- Lymphoma
- Other causes of upper airway obstruction include nasal polyposis, deviated nasal septum, rhinitis, and sinusitis (infectious, allergic, or nonallergic).

◆ Evaluation

History

History usually indicates irregular snoring with gasping or witnessed pauses and daytime somnolence. A history of behavior problems, attention deficit–hyperactivity disorder, and enuresis should be elicited. Any previous sleep studies should be reviewed. It is important to ask for any family history of known bleeding disorders, or a patient or family history of abnormal bruising or bleeding.

Physical Exam

Complete head and neck exam with focus on:

1. Assessment of the status of the middle ear (there is an increased incidence of acute OM and OME with adenoid hypertrophy)
2. Examination of the oral cavity, inspection, and grading of palatine tonsils: grade 1 (tonsils within the tonsillar pillars), grade 2 (< 50% obstruction), grade 3 (> 50% obstruction), grade 4 (where tonsils abut in the midline)
3. Exclusion of obvious craniofacial malformation or syndromic features; assess for cleft palate

Consider the direct examination of adenoid size using small flexible fiberoptic scope, especially in the cooperative child.

Imaging

Consider a lateral neck plain film to assess the size of the adenoid pad.

Labs

A complete blood count (CBC) should be done, with partial thromboplastin time (PTT) and prothrombin time/international normalized ratio (PT/INR) tests preoperatively if there is any family history or suspicion for potential coagulopathy. A platelet function test (PFA-100 analyzer) is a rapid inexpensive screen for platelet dysfunction, although it is nonspecific. A positive test should prompt von Willebrand studies and a referral to hematology.

Other Tests

Nasal endoscopy should be performed to assess the degree of nasopharyngeal obstruction. Rhinomanometry is rarely used to assess the degree of nasal obstruction.

An otherwise generally healthy child with parents who provide a good description of irregular snoring with gasping, pausing, or choking noises, and an exam that reveals obvious obstructive hypertrophy, does *not* need a polysomnography; proceed with an adenotonsillectomy. If the child has an unconvincing exam or the parents cannot provide a good history, polysomnography is helpful. If the child has other comorbidities, and *especially* if the child has neurologic problems, polysomnography is important, as central rather than obstructive apnea may be occurring. Central apnea will not be improved by an adenotonsillectomy.

Pathology

Hyperplasia of the lymphoid tissue of the adenoids and tonsils is found.

◆ Treatment Options

Medical

- Watchful waiting
- Topical nasal steroids
- Antibiotics may shrink tonsils occasionally
- Continuous positive airway pressure (CPAP) for OSA (see **Chapter 1.3**)

Surgical

The surgical procedures employed include adenoidectomy, tonsillectomy, and adenotonsillectomy. For instrumentation, most surgeons currently use the electrocautery. Also, some surgeons perform cold dissection, use a bipolar radiofrequency ablation device (coblation), or use a microdébrider for so-called intracapsular or subtotal tonsillectomy. Other devices have been introduced and marketed but are less widely used.

◆ Complications

Bleeding

The most common significant complication is postoperative hemorrhage, which occurs at a rate of 2 to 3%. The patient/family must understand risk of delayed bleeding and the need to present to the emergency department promptly if there is any bleeding. Children in rural areas should not be allowed to be more than 30 minutes from a hospital for 2 weeks.

Bleeding may be classified as intraoperative, primary (occurring within the first 24 hours), or secondary (occurring between 24 hours and 10 days). Primary tonsillar hemorrhage (0.5–2.2 per 100) often requires operative management. This is due either to a failure to control a vessel that was likely in spasm or to an unrecognized clotting problem, typically platelet dysfunction.

Secondary (after first 24 hours) tonsillar hemorrhage (0.1–3 per 100) is often managed conservatively with close observation. This is most common at ~ 1 week postoperatively.

Recurrent secondary hemorrhage should prompt a thorough hematologic evaluation and mandates admission with close observation based on concern for a possible "sentinel" bleed (**Table 8.6**).

Table 8.6 Management options for posttonsillectomy bleeding

If active bleeding, go directly to the operating room for definitive control.
If inactive or scant bleeding: Admission with 24-hr observation. If rebleed, go to operating room. If no bleeding for 24 hr, discharge.
All patients: Check hemoglobin, hematocrit, PT, PTT. Establish intravenous access, hydrate.
Possible measures to control mild/scant bleeding in emergency department: silver nitrate or other chemical cautery

Abbreviations: PT, prothrombin time; PTT, partial thromboplastin time.

Other Complications

In some studies respiratory compromise is the most frequent complication occurring in children after tonsillectomy. The risk of respiratory complications is 4.9 times higher in those who have obstructive sleep apnea than in children who do not. Early postoperative airway obstruction may require humidified oxygen, steroids, and reintubation. Chronically obstructed patients are at risk of postoperative pulmonary edema, diagnosed with physical exam and chest radiography. This may require furosemide, humidified oxygen, chest physiotherapy, positive end expiratory pressure ventilation, and reintubation.

Postoperative dehydration may require readmission. Treat with intravenous (IV) hydration, antiemetics, analgesia as needed, and parental education.

Nasal reflux is commonly mild and resolves; if severe or persistent, this may be difficult to correct. Speech therapy can be helpful.

Nasopharyngeal stenosis/scarring is uncommon and extremely difficult to correct. Prevention is best.

Airway fire: This should be extremely rare. One should check for cuff leak before operating; should not leak below 20. Communicate with the anesthesiologist, and be sure FIO_2 is kept less than 40%. If there is a fire, then immediately extubate, turn off all the oxygen, and reintubate.

◆ Outcome and Follow-Up

Close postoperative monitoring in children with OSA, given the risk of ongoing or worsening apnea in the immediate postoperative period. Following tonsillectomy, children with a positive preoperative sleep study and/or any children 3 years or younger are typically observed overnight with continuous pulse oximetry, although there is a lack of consensus on this issue.

Other measures: Liquid/soft diet for 2 weeks postoperative. Pain management is important to enable the patient to maintain adequate hydration. Although there is no consensus, a recommended combination is acetaminophen liquid up to 15 mg/kg/ dose three or four times daily, using plain oxycodone liquid 0.1 mg/kg/dose (5 mg maximum) every 4 hours as needed for breakthrough pain. Separating the acetaminophen and the opioid enables the patient to minimize opioid use in many cases. Nonsteroidal anti-inflammatory drugs (NSAIDs) are avoided out of concern for platelet inhibition, although some studies suggest this does not increase bleeding risk. Oral liquid antibiotic is used for 1 week postoperatively, usually amoxicillin if the patient is nonallergic. Intraoperatively, a single dose of dexamethasone 0.5 mg/kg IV is helpful to reduce postoperative pain and swelling; the dose is typically 4 to 12 mg.

Continued routine follow-up if conservative treatment approach adopted. Repeat sleep study may be considered 3 to 6 months postoperatively if symptoms persist. Follow-up is scheduled 3 to 6 weeks after adenotonsillectomy.

Adenotonsillectomy achieves a 90% success rate for childhood sleep-disordered breathing. Of all tonsillectomies and adenoidectomies performed in the United States each year, 75% are performed to treat sleep-disordered breathing.

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8.12 Pediatric Hearing Loss

◆ Key Features

- Age at identification has significantly decreased since the advent of newborn hearing screening.
- Early identification and treatment of hearing loss is essential to obtain an optimal outcome.
- Connexin-26–related deafness is the most common cause of inherited hearing loss.
- Acquired hearing loss often occurs within the antenatal period.
- Perinatal history is the key to identifying risk factors for both congenital and acquired hearing loss.

The onset of hearing loss in children regardless of etiology often occurs prior to the development of language. The disorders causing both congenital and acquired hearing loss range from the simple and common to the rare and complex. The list of potential diagnoses in this chapter is not exhaustive; additional syndromes with hearing loss as a minor characteristic can be found in **Chapter 8.9**. For the purpose of this chapter, we will examine hearing loss using the structure seen in **Fig. 8.3**, with the caveat that some diagnoses may fall under more than one category.

◆ Epidemiology

Hearing loss is the most common congenital sensory deficit, with an estimated incidence of 2 per 1,000 children. Likewise, acquired loss is www.konkur.in

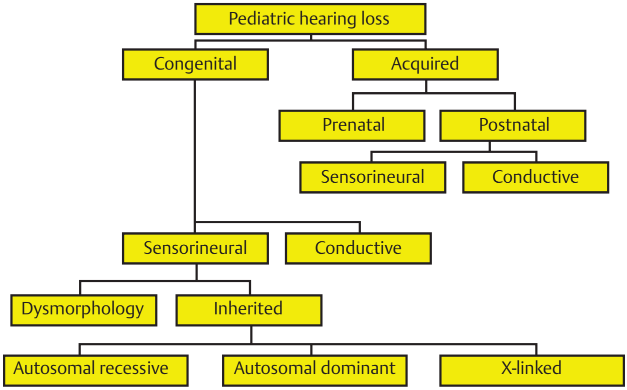

Fig. 8.3 Types of pediatric hearing loss.

exceedingly common, with most children experiencing at a minimum a transient conductive hearing loss (CHL) due to chronic serous otitis. An estimated 50% of childhood sensorineural hearing loss (SNHL) is due to genetic factors, with this proportion increasing with the ongoing detection of new mutations. Genetic causes may eventually account for a large proportion of the hearing loss currently labeled as unknown in etiology. For cases of congenital hereditary hearing loss, about two-thirds are nonsyndromic. In the setting of genetic-related deafness, most cases (70–80%) are autosomal recessive, roughly 20% are autosomal dominant, and the remainder are due to X-linked chromosomal or mitochondrial anomalies. About 50% of the cases of congenital SNHL are considered nonhereditary (i.e., acquired), and most of these are due to TORCHES (*to*xoplasmosis, *r*ubella, *c*ytomegalovirus, *h*erpes simplex *e*ncephalitis, and oto*s*yphilis) infections, sepsis, or severe prematurity.

◆ Clinical

Signs

- Speech delay or regression of speech
- Vestibular dysfunction
- Delays in ambulation
- Gait disturbances
- Otorrhea

Symptoms

- Hearing loss (may be progressive or fluctuating)
- Tinnitus
- Vertigo
- Otalgia
- Aural fullness

◆ Congenital Sensorineural Hearing Loss

Dysmorphologies

Most cochlear dysmorphologies are membranous in nature (80–90%) and not identifiable on computed tomography (CT); the remainder have a discernible bony anomaly that is identifiable on CT imaging.

Mondini Deformity

- Incomplete partition of cochlea
- Autosomal dominant
- Unilateral or bilateral hearing loss may be progressive or fluctuating (interspersed with normal hearing).
- CT: Cystic dilation of cochlea with absence of modiolus, enlarged vestibular aqueduct, semicircular canal anomalies
- Associated with Pendred's, Waardenburg's, and Treacher Collins's syndromes and branchio-oto-renal syndrome (described subsequently)

Michel Aplasia

- Autosomal dominant
- Anacusis
- CT: Absent cochlea and labyrinth, hypoplasia of petrous pyramid

Alexander Deafness

- Autosomal recessive or sporadic inheritance
- Most common inner ear aplasia
- High-frequency loss with residual low-frequency hearing
- Aplasia of cochlear duct (membranous defect)
- CT: No characteristic features
- Associated with congenital rubella and Jervell and Lange-Neilsen's, Usher's, and Waardenburg's syndromes

Scheibe Aplasia

- Autosomal recessive
- Most common inner ear dysplasia
- Partial to complete aplasia of cochlea and saccule (pars inferior) with normal semicircular canals and utricle (pars superior)
- CT: No characteristic features (membranous defect)
- Associated with Usher's and Waardenburg's syndromes

Enlarged Vestibular Aqueduct

- Most common radiologic deformity of the inner ear
- Progressive SNHL that may be sudden or stepwise decrease
- Can occur in isolation or with Mondini deformity
- Associated with progressive and/or fluctuating vestibular dysfunction and Pendred's syndrome
- Associated with perilymph gusher

Bing-Siebenmann Dysplasia

- Rare, complete dysplasia of membranous labyrinth

Inherited Disorders

Autosomal Recessive

Connexin 26

- Chromosomal mutation on 13q11
- Most common genetic cause of deafness (> 50% of recessive nonsyndromic hearing loss)
- Most commonly autosomal recessive (90 mutations identified)
- Can be autosomal dominant (nine mutations identified)
- Quantitative and qualitative defects in protein coding for gap junctions (GJB2), which are responsible for maintaining endolymphatic potential in the cochlea [K^+], caused by the gene mutation
- 35delG the most common mutation (however, > 100 mutations have been identified)

Usher's Syndrome

- Chromosome arm 14q (type I), chromosome arm 1q32 (type II)
- SNHL, retinitis pigmentosa with or without mental retardation, and cataracts

See **Table 8.7**.

Table 8.7 Classification of Usher syndrome

Type	Sensorineural hearing loss	Vestibular function	Blindness
I	Profound	Areflexia	Early adulthood
II	Moderate to severe	Normal	Midadulthood
III	Progressive	Progressive	Variable

Pendred's Syndrome

- Chromosomal mutation on 7q coding for pendrin (sulfate transporter)
- *SLC26A4* the most common mutation
- Defect in tyrosine iodination
- Severe to profound SNHL
- CT: Mondini deformity or isolated enlarged vestibular aqueduct
- Associated with euthyroid multinodular goiter in childhood

Jervell and Lange-Nielsen's Syndrome

- Chromosomal mutation on 11p15 as well as 3, 4, 7, 21
- Bilateral SNHL
- Long Q-T syndrome (treated with β-blockers)

Goldenhar's Syndrome

- Hemifacial microsomia
- Chromosomal mutation on 9
- Developmental anomalies of the first and second branchial arches affecting facial nerve, stapedius muscle, semicircular canals, and oval window
- Associated with preauricular tags, pinnae anomalies, aural atresia, facial asymmetry, colobomas, and mental retardation

Autosomal Dominant

Waardenburg's Syndrome

- Chromosomal mutation on 2q and 14q
- Unilateral or bilateral SNHL and/or vestibular dysfunction
- Associated with pigment anomalies (white forelock, heterochromia iridum [different colored irises]), telecanthus, synophrys (unibrow), skeletal anomalies, Hirschsprung's disease

Stickler's Syndrome

- Progressive arthroophthalmopathy
- Chromosomal mutation on 6p and 12q
- Variable expression
- Progressive SNHL and/or conductive loss (eustachian tube dysfunction due to cleft palate)
- Associated with myopia and/or retinal detachment, cataracts, Pierre Robin's sequence, marfanoid habitus, joint hypermobilization, and early arthritis

Treacher Collins's Syndrome

- Mandibulofacial dysostosis
- Chromosomal mutation on 5
- Conductive hearing loss: EAC atresia, ossicular malformations, replacement of tympanic membrane with bony plate
- Sensorineural hearing loss: widened cochlear aqueduct
- Associated with aberrant facial nerve, auricular anomalies, preauricular fistulas, mandibular hypoplasia, coloboma, palate defects, and downward-slanting palpebral fissures

Branchio-Oto-Renal Syndrome

- Chromosome mutation on 8q
- *EYA1* and *SIX1* mutations have been identified.
- Developmental anomalies of the branchial arches and kidneys
- Variable SNHL
- CT: Possible Mondini deformity
- Associated with auricular deformities, preauricular pits, fistulas, tags, mild renal dysplasia progressing to agenesis

Neurofibromatosis

- Neurofibromatosis type 1 (NF1): Long arm of chromosome 17; 5% risk of unilateral vestibular schwannoma
- Neurofibromatosis type 2 (NF2): Long arm of chromosome 22; frequently bilateral vestibular schwannomas, multiple meningiomas
- Progressive profound SNHL due to acoustic neuroma
- Associated with café-au-lait spots, groin/axillary freckling, cutaneous, central nervous system (CNS) and peripheral nerve neurofibromas, Lisch nodules (eye hamartomas), and pheochromocytomas

Apert's Syndrome

- Acrocephalosyndactyly
- Chromosomal mutation on 10
- Autosomal dominant or sporadic inheritance
- CHL due to stapes fixation
- CT: patent cochlear aqueduct, large subarcuate fossa
- Associated with syndactyly, midface anomalies including hypertelorism, proptosis, saddle nose deformity, high-arched palate, trapezoid mouth, and craniofacial dysostosis

Crouzon's Syndrome

- Craniofacial dysostosis
- Chromosomal mutation on 10q; CHL to aural atresia and ossicular deformities
- Associated with cranial synostosis, small maxilla, hypertelorism, exophthalmos, prognathic mandible, and short upper lip

Osteogenesis Imperfecta

- Chromosomal mutation on 5p or 17q
- Typically autosomal dominant inheritance, although a familial recessive form has been described
- Disorder of type I collagen (quantitative and qualitative collagen defect); progressive conductive, mixed, or sensorineural hearing loss
- Associated with hypermobility of joints and ligaments, bone fragility, and blue sclera

X-Linked

Alport's Syndrome

- Chromosome Xq, some autosomal dominant mutations in chromosome 2q
- Abnormality of type IV collagen formation in basement membrane
- Progressive SNHL due to degeneration of organ of Corti and stria vascularis within first decade
- Associated with progressive nephritis (hematuria, proteinuria, chronic glomerulonephritis, uremia), myopia, and cataracts

- *Oto-Palato-Digital Syndrome (see also* **Chapter 8.9**)
- CHL due to ossicular malformations
- Associated with cleft palate, broad fingers and toes, hypertelorism, mental retardation, and short stature

Norrie's Disease

- Chromosomal mutation on Xp11
- Progressive SNHL (in one third of patients; in second or third decade)
- Rapidly progressive blindness (bilateral pseudoglioma, exudative vitreo-retinopathy, and ocular degeneration)

Wildervanck's Syndrome

- Dominant X-linked
- SNHL (one third of cases)
- Associated with CN VI paralysis and Klippel-Feil syndrome (fusion of cervical vertebrae)

X-Linked Hearing Loss

- Chromosomal mutation on Xq
- Rare syndrome with mixed hearing loss due to stapes fixation and perilymph gushers

◆ Congenital Conductive Hearing Loss

Atresia

Atresia may be unilateral or bilateral.

Syndromic

Treacher Collins's syndrome, Apert's syndrome, Crouzon's syndrome, oto-palato-digital syndrome, osteogenesis imperfecta, X-linked hearing loss (see foregoing syndrome descriptions)

◆ Acquired Hearing Loss

Acquired hearing loss is often due to prenatal or perinatal insult, such as infection.

Acquired Prenatal

Infection *before* birth results in hearing loss.

Rubella

- Hair cell loss, atrophy of the organ of Corti, thrombosis of stria vascularis
- Severe to profound SNHL and/or delayed endolymphatic hydrops
- Associated with mental retardation, lower limb deformities, anemia, cataracts, cardiac malformations, microcephaly, and thrombocytopenia

Syphilis

- *Treponema pallidum*
- Often fatal
- Profound SNHL (within first 2 years, or in second or third decade)
- Hennebert's sign (nystagmus with fluctuating pressure on external auditory canal [EAC] in the presence of an intact tympanic membrane—in delayed congenital syphilis)
- Delayed endolymphatic hydrops
- Penicillin for acquired infection and/or corticosteroids for SNHL

Cytomegalovirus (CMV)

- Mild to profound SNHL
- May be unilateral and/or progressive
- Associated with hemolytic anemia, microcephaly, mental retardation, hepatosplenomegaly, cerebral calcifications, and jaundice

Hypoxia

- SNHL and/or auditory neuropathy
- Risk factors include birth anoxia, postnatal hypoxia and/or ventilatory support, and extracorporeal membrane oxygenation (ECMO).

Hyperbilirubinemia

- SNHL and/or auditory neuropathy

Acquired Postnatal

Infection or injury *after* birth results in hearing loss.

Sensorineural Loss

Meningitis

- SNHL typically bilateral, permanent, profound to severe (15–20% of cases)
- Onset of SNHL early in this disease
- Bacterial etiology: *Haemophilus influenzae* (most likely to cause hearing loss), *Streptococcus pneumoniae* (most common cause of meningitis in children; most likely to cause labyrinthine ossification)
- Access of bacteria and their toxins to inner ear likely via the cochlear aqueduct and the internal auditory canal (IAC)
- Possible role of inflammatory, hypoxic injury to neural elements
- Repeated auditory brainstem response assessment starting early in the course of disease
- CT: May have labyrinthine ossification. Lateral semicircular canal is often first to ossify. If child medically stable and fulfills audiologic candidacy, cochlear implantation should occur immediately ($\leq$ 6 weeks) if scan shows fibrosis, because once the cochlea has ossified, implantation then requires cochlear drill-out, allowing only a partial electrode, portending a poorer outcome.

Trauma

- Possible temporal bone fracture
- SNHL due to fracture through otic capsule, stapes subluxation with perilymphatic leak, or excessive displacement and shearing of basilar membrane (leads to a high-frequency loss)
- CHL due to soft tissue and debris in the EAC, tympanic membrane perforation (20–50%), hemotympanum (20–65%), ossicular dislocation or injury (incus most commonly dislocated, more commonly associated with longitudinal fracture)
- Associated with facial nerve injury, vertigo, and posttraumatic benign paroxysmal positional vertigo

Auditory Neuropathy/Dyssynchrony

- Disorder of central processing
- Poor discrimination scores out of keeping with pure tone thresholds (absent auditory brainstem response, normal otoacoustic emissions [OAEs])
- Risk factors include hyperbilirubinemia, hypoxia or mechanical ventilation, low birth weight, congenital anomalies of the CNS, Stevens-Johnson‘s syndrome
- May accompany neurologic diagnosis (Friedrich’s ataxia, Ehlers-Danlos’s syndrome, Charcot-Marie-Tooth’s syndrome)
- Most commonly acquired, although inherited variants exist
- May perform well with cochlear implantation due to suprathreshold synchronous stimulation it supplies to auditory nerve

Conductive Loss

- Cholesteatoma
  - Ossicular erosion leading to maximal conductive loss
- Tympanic membrane perforation
  - CHL (30–50 dB)
  - Successful repair may not improve hearing.
  - Perforation due to acute OM, chronic OM, barotraumas, complication of tympanostomy tube insertion or middle ear surgery
  - Should obey water precautions, particularly in bath water and lake water
- Chronic serous otitis
- Trauma
  - See full description under sensorineural loss.
- Ear foreign body
  - See **Chapter 3.1.4**.

◆ Evaluation

Physical Exam

See **Chapter 3.4.2** for pediatric audiologic assessments. A complete head and neck exam should include otoscopy, pneumatoscopy, Rinne and Weber tuning fork tests if age-appropriate, and inspection of the auricle and preauricular skin for pits or tags. A clinical examination of the vestibular system appropriate for the child's age should be undertaken, as well as a general examination to identify syndromic features (see preceding discussions for differential diagnosis for characteristic syndromic features) (**Table 8.8**).

Table 8.8 Suggested evaluation for congenital sensorineural hearing loss

Syndromic • Refer to geneticist. • Consider: electrocardiography, urinalysis • Other imaging as directed by specific syndrome
Nonsyndromic • Tests for possible acquired causes: • RPR/FTA-ABS • TORCH IgM assay • Genetic testing: testing for known common connexin mutations; various genetic panels (such as CapitalBioMiamiOtoArray; MiamiOtoGenes) others
All • CT of temporal bone • Referral to ophthalmology • Serial hearing testing to follow possible progression

Abbreviations: CT, computed tomography; IgM, immunoglobulin M; RPR/FTA-ABS, rapid plasma reagin/fluorescent treponemal antibody absorbed; TORCH, toxoplasmosis, rubella, cytomegalovirus, herpes simplex encephalitis, and otosyphilis.

Imaging

High-resolution CT of the temporal bones will identify cochleovestibular anomalies. Consider magnetic resonance imaging (MRI) with attention to the IAC to examine for the presence and size of the auditory nerve, white matter changes due to sequelae of prematurity, CMV, hyperbilirubinemia, or in cases of neurofibromatosis.

Labs

- Pendred's syndrome: thyroid-stimulating hormone (TSH) test
- Alport's, branchio-oto-renal syndrome: urinalysis, BUN (blood urea nitrogen), creatinine
- TORCH: IgM assay (toxoplasmosis, syphilis, rubella, CMV, herpes simplex)
- Rubella: Viral culture of amniotic fluid, urine, and throat
- Syphilis: fluorescent treponemal antibody absorbed (FTA-ABS), Venereal Disease Research Laboratory (VDRL) test

Other Tests

- Repeated audiometric assessment with tympanometry appropriate for age (**Table 8.9**)
- Assessment of visual acuity due to exponential increase in morbidity in presence of dual sensory impairment
- Genetic evaluation, counseling: connexin-26, syndromic and/or inherited loss
- Jervell and Lange-Nielsen's syndromes: Electrocardiogram to detect prolonged Q-T interval
- Pendred's syndrome: thyroid ultrasound
- Branchio-oto-renal syndrome: renal ultrasound, pyelogram

Table 8.9 Audiometric assessment by age

Birth to 6 months • Behavioral observation audiometry • Otoacoustic emission • Auditory brainstem response
6 months to 3 years • Visual response audiometry • Otoacoustic emission • Auditory brainstem response
3–6 years • Conventional play audiometry
> 6 years • Standard audiometry

◆ Treatment Options

Medical

Appropriate and timely fitting of amplification (by 6 months if identified by newborn hearing screening) with a frequency modulation (FM) system or tactile aids. An auditory verbal therapist or a total communication program, including American Sign Language (ASL) where appropriate, should be considered. Liaise with the school, teachers, and the educational audiologist.

Surgical

- Tympanostomy tubes: Mixed loss due to chronic serous OM
- Cochlear implantation for profound bilateral SNHL (see **Chapter 3.5.4**)
- Bone-anchored hearing aid for aural atresia, chronic OM (see **Chapter 3.5.5**)
- Surgical correction of atretic ear and canal
- Middle ear exploration and/or ossiculoplasty, tympanoplasty for suspected congenital or acquired conductive loss
- Auditory brainstem implant (see **Chapter 3.5.5**) for:

- Bilateral acoustic neuroma in NF2
- Cochlear or auditory nerve agenesis (i.e., Michel aplasia)

◆ Complications

Treat TM perforation due to chronic OM or as a complication of tympanostomy tube insertion with delayed tympanoplasty once a dry ear is established. Treat OM aggressively with antibiotics and tympanostomy tubes in children following cochlear implantation due to risk of meningitis.

◆ Outcome and Follow-Up

Postoperatively, apply analgesia and/or systemic antibiotics where the auditory prosthesis has been placed. Children receiving cochlear implantation should receive pneumococcal and meningitis vaccination as prophylaxis prior to implantation. Ongoing and repeated audiologic assessment should be maintained with attention to the child's speech and language development.

With a child with tympanostomy tubes, follow up every 6 months until the tubes have extruded and the myringotomy site has closed without reaccumulation of effusion.

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